

RKDF University

Program Outcomes, Program Specific Outcomes and Course Outcomes

Faculty of Computer Application

Bachelor of Computer Application (BCA)

Programme Outcome (POs)

- PO 1:** Acquire skills and information not only about Computer and Information Technology but also in communication, organization and management.
- PO 2:** They are well equipped with the skills of engineering approach in software development.
- PO 3:** Get to learn programming languages such as C, C++, HTML, SQL, DBMS, and Networking etc.
- PO 4:** Information about various computer applications and latest developments in IT and communication systems is also provided. They develop an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
- PO 5:** apply knowledge of computing and mathematics appropriate to the discipline.
- PO 6:** identify, formulate, and develop solutions to computational challenges.
- PO 7:** design, implement, and evaluate a computational system to meet desired needs within realistic constraints.
- PO 8:** function effectively on teams to accomplish shared computing design, evaluation, or implementation goals.
- PO 9:** Ability to work in team and build leadership qualities.
- PO 10:** Importance of time in completing the projects on time.
- PO 11:** Will be well equipped with thorough knowledge of various softwares.
- PO 12:** Will be familiar with working with various operating system platforms.
- PO 13:** Understand the professional, ethical, legal, security, and social issues and responsibilities in computing profession.
- PO 14:** Analyse impacts of computing on individuals, organizations, and society.

PO 15: Recognition of the need for and ability to engage in continuing professional development.

PO 16: Will be able to choose appropriate techniques, skills, and tools necessary for

PO 17: Designing of correct models in the construction of software systems of varying complexity.

PROGRAMME SPECIFIC OUTCOME (PSOs)

PSO1: Equip themselves to potentially rich & employable field of computer applications.

PSO 2: Pursue higher studies in the area of Computer Science/Applications.

PSO 3: Take up self-employment in Indian & global software market.

PSO 4: Meet the requirements of the Industrial standards.

COURSE OUTCOME

Semester – I

English Language FC-101/1

C 01: In this course students will read and understand about the rich classical texts from Indian literatures written in Hindi as well as Indian literatures written in Urdu, in translated versions

C 02: Upon the completion of course the students will be able trace the nature of influence that the classical texts have on modern English literatures

C 03: Know about various innovative ways of using English language in verbal and non-verbal communications.

C 04: Write clearly, effectively, and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.

C 05: Think about the relation between language and literature

Development of Entrepreneurship FC-101/2

C 01: In this course students will learn and understand the meaning of entrepreneurship, types and functions of an entrepreneur, sequenced planning and guiding capacity explanation.

C 02: Upon the completion of the course the students will be able to discern distinct entrepreneurial traits

C 03: They will understand the systematic process to select and screen a business idea

C 04: Students will be able to design strategies for successful implementation of ideas

C 05: They will be able to prepare and analyse Project Report and interpret the same.

Computer Fundamentals BC-102

CO1: To acquire the knowledge of Computer System and its Generations

CO2: To acquire the knowledge of Input devices, Output devices and Storage devices

CO3: To acquire the knowledge of Programming Languages its advantage and disadvantage

CO4: To understand the concept of Software and Operating Systems with application

CO5: To acquire the knowledge of Algorithms and Flow Charts of a programme

PC Software BC-103

CO1: To acquire the knowledge of Disk Operating System with Internal Command and external Command

CO2: To acquire the knowledge of Installation of MS Office with its different versions

CO3: To acquire the knowledge of Word Processing and its advantages

CO4: To acquire the knowledge of Spread sheet with its utility in Excel

CO5: To acquire the knowledge of Making a Formula in Excel

Digital Computer Electronics BC-104

CO1: To acquire the knowledge of Number System and its conversion

CO2: To acquire the knowledge of Logic gates and their Truth Tables

CO3: To understand the fundamental of Primary and secondary memory

CO4: To acquire the knowledge of CPU organization and its types

CO5: To understand the Data transfer schemes with its applications

Communication and Management BC-105

CO1: To acquire the knowledge and understanding of Communication and its important to managers

- CO2:** To acquire the knowledge Effective Listening
- CO3:** To acquire the knowledge of Verbal and Non-verbal Communication
- CO4:** To develop the understanding of Business Communication and its impact on real life.
- CO5:** To acquire the knowledge of Drafting an Official Letters

Semester – II

Hindi Sanrachna FC 201/1

C 01: पाठ्यक्रम पूरा करने के बाद, एक शिक्षार्थी को हिंदी भाषा के बारे में एक उन्नत समझ होगी

C 02: पाठ्यक्रम पूरा करने के बाद, लेख लिखने के साथ-साथ अधिग्रहण करने के लिए एक शिक्षार्थी, वक्तृत्व में और हिंदी भाषा में अध्ययन में निपुणता होगी

C 03: हिंदी विद्वानों को बढ़ावा देने के लिए और उन्हें प्रासंगिक लिखने और अनुवाद करने के लिए प्रोत्साहित करने के लिए

C 04: छात्रों को उनकी समृद्ध सांस्कृतिक, नैतिक, साहित्यिक और वैज्ञानिक विरासत और के लिए जागृत करना

Environment FC-201/2

C 01: The subject facilitate the students' understanding of complex environmental issues from a problem-oriented, interdisciplinary perspective

C 02: Understand environmental problems and ways of addressing them, including interactions across local to global scales.

C 03: Upon the completion of the course the students will be able to appreciate the ethical, cross-cultural, context of environmental issues and the links between human and natural systems.

C 04: They will reflect critically about their roles and identities as citizens, consumers and environmental actors in a complex, interconnected world.

Business Mathematics BC-202

CO1: To acquire the knowledge of basics of Trigonometry

CO2: To learn the theory of Indices and the type of Matrices

CO3: To understand the Frequency Distribution with Measure of Central Tendency

CO4: To acquire the knowledge of Determinant and Matrices

CO5: To understand the meaning of Limits & Continuity with examples

Desktop Publishing (DTP) BC-203

CO1: To develop the basic understanding of DTP Package

CO2: To acquire the knowledge of PageMaker and Preparation of Document Using DTP Package

CO3: To acquire the knowledge of various types of Printers used in DTP.

CO4: To understand the commercial DTP System that is available in market

CO5: To acquire the knowledge of Import & Export of Documents in Word Processors

Programming in 'C' BC-204

CO1: To acquire the knowledge of basics of C language

CO2: To acquire the knowledge of Decision control structure and arithmetical operators

CO3: To acquire the knowledge of concept of loops with examples

CO4: To acquire the knowledge of development of C programs using Arrays & functions

CO5: To acquire the knowledge of how elements are stored in array

Accounting Management BC-205

CO1: To understand the basic concept of Accounting

CO2: To understanding the fundamentals of insurance method, Sinking Fund Method

CO3: To develop basic knowledge of Cash Book, Bills of Exchange

CO4: To develop a basic understanding of profit and loss account

CO5: To acquire the knowledge of Financial Statements and its analysis

Semester – III

Values & Spirituality FC-301/1

- C 01:** The course aims to teach and inculcate the importance of value based living among students and give them a deeper understanding about the purpose of life.
- C 02:** Upon the completion of the course the students will understand the importance of value based living.
- C 03:** Students will understand and start applying the essential steps to become good leaders and emerge as responsible citizens with clear conviction to practice values and ethics in life.
- C 04:** Students will become value based professionals and will contribute in building a healthy nation.

Environment FC-301/2

- C 01:** The course aims at providing students with a broad interdisciplinary liberal arts framework for understanding the relationship between humans and their environment;
- C 02:** Students will contribute to and facilitate interdisciplinary research and problem solving, through independent and collaborative work
- C 03:** Upon the completion of the course students will be able to demonstrate an integrative approach to environmental issues with a focus on sustainability;
- C 04:** Communicate complex environmental information to both technical and non-technical audiences;
- C 05:** Reflect critically on their roles, responsibilities, and identities as citizens, consumers and environmental actors in a complex, interconnected world.

Operating System BC-302

- CO1:** To understand the basics of Operating Systems and its Components
- CO2:** To gain the knowledge of CPU scheduling and algorithms
- CO3:** To understand the concept of memory management techniques
- CO4:** To acquire the basic knowledge of file management system & directory structure
- CO5:** To acquire the basic knowledge of Deadlocks with the techniques of avoidance, detection, prevention and recovery

Database Management and Design BC-303

CO1: To Gain the knowledge of Database Theory, its advantage and disadvantages

CO2: To acquire the knowledge of Relational Database and the concept of Normalization

CO3: To acquire the knowledge of Relational Algebra & Fundamental Operations with SQL

CO4: To formulate SQL queries on data using basic DDL, DML and DCL commands.

CO5: To acquire the knowledge of PL/SQL programming for simple applications.

Object Oriented Programming Using C++ BC-304

CO1: To Understand the basic concepts of object-oriented programming

CO2: To acquire the knowledge of Operators in C++

CO3: To Gain the knowledge of Constructors and Destructors with its application

CO4: To acquire the knowledge of Inheritance property in C++ and its utility in programming

CO5: To acquire the knowledge of Exception handling in C++

Organizational Behaviour BC-305

CO1: To acquire the basic knowledge of Psychology with observation

CO2: To study Sensation & Perception with Feelings and Emotions

CO3: To Gain knowledge of theories of Motivation and theory of Organization

CO4: To acquire the basic knowledge Theories of Leadership

CO5: To Gain knowledge of Communication and its usefulness

Semester – IV

English Language FC-401/1

C 01: The course aims at exposing students to the examples of prose and poetry in English so that they realize the beauty and communicative power of English

C 02: Upon the completion of the course the students will develop the ability to appreciate ideas and think critically

C 03: On successful completion of the Programme, the students will be accurate both in oral and written communication as they will be strong in Writing skills and its usage.

C 04: They can express a thorough command of English and its linguistic Structures.

Development of Entrepreneurship FC-401/2

C 01: The students develop and can systematically apply an entrepreneurial way of thinking that will allow them to identify and create business opportunities that may be commercialized successfully.

C 02: After the completion of the course, the students will be able to have the ability to discern distinct entrepreneurial traits

C 03: Know the parameters to assess opportunities and constraints for new business ideas

C 04: Understand the systematic process to select and screen a business idea and design strategies for successful implementation of ideas

C 05: The students will be able to write a business plan

Computer Oriented Numerical Methods BC-402

CO1: To acquire the basic knowledge of Algebraic Equations with different method Newton Raphson method, Regula-Falsi method and Successive Bisection method and its application

CO2: To acquire the basic knowledge of Interpolation & Extrapolation specially computation of missing value; and Bessel's Central difference formula

CO3: To acquire the basic knowledge of Integration & Differentiation and its application

CO4: To acquire the basic knowledge for the solution of ordinary differential & Integral Methods: Euler's, Picard's and Taylor's series methods.

CO5: To understand the meaning of Correlation, Regression and Lines of Regression with their application

System Analysis & Design BC-403

CO1: To acquire the elementary knowledge of System Analysis & Design its type and System Development Life Cycle

CO2: To acquire the elementary knowledge structured analysis and learn the Steps for Feasibility Study and for making the Feasibility Report.

- CO3:** To acquire the elementary knowledge System Testing, Why Testing is necessary.
- CO4:** To acquire the elementary knowledge Software Maintenance and how to reducing maintenance costs
- CO5:** To Gain the knowledge of System Security, Data Security, Threats that required for Risk Analysis

Web Design BC-404

- CO1:** To acquire the elementary knowledge of Internet, URL and Web Server
- CO2:** To acquire the elementary knowledge for formatting the HTML documents layout
- CO3:** To understand the Advanced Layout: Tables, Frames, Layers and their attributes
- CO4:** To acquire the elementary knowledge of style sheet and its properties
- CO5:** To acquire the elementary knowledge DHTML, XML and Web publishing

Data Structure & Algorithms BC-405

- CO1:** To acquire the elementary knowledge of Data Structure, its types and utility of array
- CO2:** To Gain the knowledge of Stack and learn the program of Queues & its application
- CO3:** To acquire the elementary knowledge of Linked list and Linked implementation of Stacks and queues
- CO4:** To understand the basic knowledge of binary tree and Insertion & Deletion algorithms with its utility
- CO5:** To acquire the elementary knowledge of Selection Sort, Bubble Sort method & its algorithm

Semester – V

Values & Spirituality FC-501/1

- C 01:** The course aims to teach and inculcate the importance of value based living among students and give them a deeper understanding about the purpose of life.

- C 02:** Upon the completion of the course the students will understand the importance of value based living.
- C 03:** Students will understand and start applying the essential steps to become good leaders and emerge as responsible citizens with clear conviction to practice values and ethics in life.
- C 04:** Students will become value based professionals and will contribute in building a healthy nation.

Basic Computer Information Technology-I FC-501/2

- CO1:** To acquire the elementary knowledge of the development of Computer system and its generation
- CO2:** To acquire the elementary knowledge of ALU, Input/ Output and kind of memory RAM, ROM, EPROM and PROM.
- CO3:** To gain the elementary knowledge of the memory management and its application
- CO4:** To understand the storage devices and its capacity of in term of storages
- CO5:** To acquire the elementary knowledge operating systems, it's functioning and types. Also learn the basic commands of dos & Windows operating System.

Basic Computer Information Technology-I FC 501/2

- C 01:** After studying this course the students will understand the fundamental hardware components that make up a computer's hardware and the role of each of these components.
- C 02:** Have an understanding about an operating system and an application program, and what each is used for in a computer
- C 03:** Demonstrate a basic understanding of computer hardware and software.
- C 04:** Students will apply the computer concepts and skills learned to solve business problems
- C 05:** Students will understand the basic concepts and terminology related to computer technology

Data Communication and Networking BC-502

- CO1:** To acquire the basic knowledge of Networking and Network Topologies with Different Transmission Media

CO2: To Gain the elementary knowledge of OSI Reference Model and IEEE 802 standards

CO3: To acquire the knowledge of IP addressing MPLS and ATM applications

CO4: To acquire the knowledge of Cryptography and various Cryptography algorithms

Visual Basic BC-503

CO1: To acquire the basic knowledge of Visual BASIC, VB Controls and Operators

CO2: To acquire the basic knowledge of Msg Box, Input Box statement

CO3: To acquire the basic knowledge of Data Handling – using Grids, Validation and filtering record set.

CO4: To acquire the basic knowledge of SQL and its Components of SQL, DDL, DML

CO5: To gain the knowledge of Data Constraints & Business rule constraints

Marketing Management BC-504

CO1: To acquire the basic knowledge of marketing research and information system

CO2: To understand the Product management and the advertising management

CO3: To gain the knowledge of Marketing strategy and marketing performance of users

CO4: To Understand the Global marketing, customer service and rural marketing with some strategies

CO5: To learn Case studies in market management with its work plan.

Management Information System BC-505

CO1: To acquire the basic knowledge of hardware components and its application in problem solving

CO2: To develop the understanding of Geographic information system and Marketing, HRM

CO3: To acquire the basic knowledge of Decision Support System

CO4: To Gain the knowledge of Quality control and Quality assurance

Hindi FC 601/1

C 01: पाठ्यक्रम पूरा करने के बाद, एक शिक्षार्थी को हिंदी भाषा के बारे में एक उन्नत समझ होगी

C 02: पाठ्यक्रम पूरा करने के बाद, लेख लिखने के साथ-साथ अधिग्रहण करने के लिए एक शिक्षार्थी, वक्तृत्व में और हिंदी भाषा में अध्ययन में निपुणता होगी

C 03:

हिंदी विद्वानों को बढ़ावा देने के लिए और उन्हें प्रासंगिक लिखने और अनुवाद करने के लिए प्रोत्साहित करने के लिए

C 04: छात्रों को उनकी समृद्ध सांस्कृतिक, नैतिक, साहित्यिक और वैज्ञानिक विरासत और के लिए जागृत करना

Basic Computer Information Technology-II FC-601/2

C 01: After studying this course the students will understand the fundamental hardware components that make up a computer's hardware and the role of each of these components.

C 02: Have an understanding about the various programmes in Ms Office like Ms Word, Ms PowerPoint and Ms Excel.

C 03: Demonstrate a basic understanding of computer hardware and software.

C 04: Students will apply the computer concepts and skills learned to solve business problems

C 05: Students will understand the basic concepts and terminology related to computer technology

Software Engineering BC-602

CO1: To acquire the basic knowledge of Software Engineering problem and approach – phased at the time of development process

CO2: To Gain the knowledge of Software Specification and its Validation at the time of software development

CO3: To learn the Software Configuration management plans and risk management

CO4: To understand the Software Quality Assurance (SQA) with Software Reviews and Software Quality factors

Java BC-603

CO1: To acquire the basic knowledge of primitive data type-integer, Short, byte, float and double and its use in programming

CO2: To acquire the basic knowledge of array and also learn string and its application in program

CO3: To understand the initialization by constructors in JAVA

CO4: To Gain the knowledge of HTML Tags & Applets and its application in JAVA

Entrepreneurship BC-604

CO1: To acquire the basic knowledge of Entrepreneur and Role of enterprises in Economic Development.

CO2: To acquire the basic knowledge of various Steps in Project Formulation and Project Evaluation in Commercial and Legal Aspects

CO3: To gain the fundamental knowledge of Common Errors in Project Formulation

CO4: To acquire the basic knowledge of NSIC, SSIB, SSIDC& SISI

RKDF University

Program Outcomes, Program Specific Outcomes and Course Outcomes

Faculty of Computer Application

Master of Computer Application (MCA)

Programme Outcomes

- PO1:** Provide students with knowledge, general competence, and analytical skills in Computer Science on an advanced level.
- PO2:** Prepare them for academics, industry, and research.
- PO 3:** Provide hands-on experience to apply computing skills in all other fields of study like Mathematics, Geography, Bio Sciences, Physics, Chemistry, Linguistics, Music, Medical Sciences etc.

Programme Specific Outcomes

- PSO 1:** Become technology-oriented with the knowledge and ability to develop creative solutions, and better understand the effects of future developments of computer systems and technology on people and society as a whole.
- PSO 2:** Acquire some development experience within a specific field of Computer Science, through project work.
- PSO 3:** Gain ability to apply knowledge of Computer Science to the real-world issues.
- PSO 4:** Get familiar with current research trends in various fields of Computer Science.
- PSO 5:** Use creativity, critical thinking, analyses and research skills.
- PSO 6:** Get prepared for placement by developing personality and soft skills.
- PSO 7:** Gain ability to communicate scientific information in a clear and concise manner.
- PSO 8:** Build up programming, analytical and logical thinking abilities.
- PSO 9:** Know the recent developments in IT, future possibilities and limitations, and understand the value of lifelong learning.

PSO 10: Get an ability to participate in debates, discussions in the society constructively.

PSO 11: Gain management skills to cater the corporate sector.

COURSE OUTCOME

Semester – I

MCA-101 Software Engineering

CO1. Provide fundamental knowledge of software engineering

CO2. To help students to develop skills that will enable them to construct software of high quality software that is reliable, and that is reasonably easy to understand.

CO3. Overall system development life cycles (SLDC) provide students to understand the overall system to project according their need.

MCA - 102 Programming with C & C++

CO1. The course aims to provide exposure to problem-solving through programming.

CO2. It aims to train the students to the basic concepts of the C & C++-programming language.

CO3. This course involves a lab component which is designed to give the student hands-on experience with the concepts.

CO4. Implement several programs in languages other than the one emphasized in the core curriculum

CO5. Understand design/implementation issues involved with variable allocation and binding, control flow, types, subroutines, parameter passing

MCA -103 Theory of Computation

CO1. To learn fundamentals of Regular and Context Free Grammars and Languages

CO2. To understand the relation between Regular Language and Finite Automata and machines.

CO3. To learn how to design Automata's and machines as Acceptors, Verifiers and Translators.

CO4. To understand the relation between Contexts free Languages, PDA and TM.

CO5. To learn how to design PDA as acceptor and TM as Calculators.

MCA -104 Computer Networks

- CO1.** Study the basic taxonomy and terminology of the computer networking and enumerate the layers of OSI model and TCP/IP model.
- CO2.** Acquire knowledge of Application layer and Presentation layer paradigms and protocols.
- CO3.** Study Session layer design issues, Transport layer services, and protocols.
- CO4.** Gain core knowledge of Network layer routing protocols and IP addressing.
- CO5.** Study data link layer concepts, design issues, and protocols.
- CO6.** Read the fundamentals and basics of Physical layer, and will apply them in real time applications.

MCA -105 Data Base Management System

- CO1.** To understand the different issues involved in the design and implementation of a database system.
- CO2.** To study the physical and logical database designs, database modeling, relational, hierarchical, and network models
- CO3.** To understand and use data manipulation language to query, update, and manage a database
- CO4.** To develop an understanding of essential DBMS concepts such as: database security, integrity, concurrency, distributed database, and intelligent database, Client/Server (Database Server), Data Warehousing.
- CO5.** To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.

Semester – II

MCA-201 Artificial Intelligence & Applications

- CO1.** Gain a historical perspective of AI and its foundations.
- CO2.** Become familiar with basic principles of AI toward problem solving, inference, perception, knowledge representation, and learning.
- CO3.** Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.

- CO4.** Experience AI development tools such as an „AI language“, expert system shell, and/or data mining tool.
- CO5.** Experiment with a machine learning model for simulation and analysis.
- CO6.** Explore the current scope, potential, limitations, and implications of intelligent systems.

MCA-202 Mobile Computing

- CO1.** This course introduces the basic concepts and principles in mobile computing.
- CO2.** This includes the major techniques involved, and networks & systems issues for the design and implementation of mobile computing systems and applications.
- CO3.** This course also provides an opportunity for students to understand the key components and technologies involved and to gain hands-on experiences in building mobile applications.

MCA -203 Statistics for Computer Application

- CO1.** The objective of this course is to familiarize the students with Statistical technique.
- CO2.** To develop the students ability to deal with numerical and quantitative issues in decision Making.
- CO3.** To enable the use of statistical, graphical and algebraic techniques wherever relevant.
- CO4.** To have a proper understanding of Statistical applications in Computer Science.

MCA-204 Design and Analysis of Algorithms

- CO1.** Analyze the asymptotic performance of algorithms.
- CO2.** Write rigorous correctness proofs for algorithms.
- CO3.** Demonstrate a familiarity with major algorithms and data structures.
- CO4.** Apply important algorithmic design paradigms and methods of analysis.
- CO5.** Synthesize efficient algorithms in common engineering design situations.

MCA-205 Java Programming & Technologies

- CO1.** Understand fundamentals of programming such as variables, conditional and iterative execution, methods, etc.
- CO2.** Understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
- CO3.** Be aware of the important topics and principles of software development.
- CO4.** Have the ability to write a computer program to solve specified problems.
- CO5.** Be able to use the Java SDK environment to create, debug and run simple Java programs.

Semester – III

MCA-301 Data Warehousing and Mining

- CO1.** To identify the scope and essentiality of Data Warehousing and Mining.
- CO2.** To analyze data, choose relevant models and algorithms for respective applications.
- CO3.** To study spatial and web data mining.
- CO4.** To develop research interest towards advances in data mining.

MCA-302 Unix and Shell Programming

- CO1.** To provide introduction to UNIX Operating System and its File System.
- CO2.** To gain an understanding of important aspects related to the SHELL and the process.
- CO3.** To provide a comprehensive introduction to SHELL programming, services and utilities.

MCA-303 Cloud Computing

- CO1.** Basics of cloud computing.
- CO2.** Key concepts of virtualization.
- CO3.** Different Cloud Computing services.
- CO4.** Cloud Implementation, Programming and Mobile cloud computing.
- CO5.** Key components of Amazon Web Services

CO6. Cloud Backup and solutions.

MCA-304 .Net Framework Technology

CO1: This course is designed to provide the knowledge of Dot Net Frameworks along with C#.

MCA-305 Elective I: EI(a) : Distributed Systems

CO1: This course provides an introduction to the fundamentals of distributed computer systems.

MCA-305 Elective I: EI(b) : Embedded Systems

CO1. Develop an understanding of the technologies behind the embedded computing systems.

CO2. To introduce students to the design issues of embedded systems.

CO3. Enable students to analyze and develop software programs for embedded systems.

MCA-305 Elective I: EI(c): Network Security

CO1. To understand basics of Cryptography and Network Security.

CO2. To be able to secure a message over insecure channel by various means.

CO3. To learn about how to maintain the Confidentiality, Integrity and Availability of a data.

CO4. To understand various protocols for network security to protect against the threats in the networks.

MCA-305 Elective I: EI(f): A WEB TECHNOLOGY AND E-COMMERCE

CO1. Be familiar with web application development software tools and environments currently available on the market.

CO2. Teach the concepts, principles and methods of web engineering.

CO3. Build web applications that are scalable, flexible to modify and easy to manage.

MCA-305 Elective I: EI(g) :Optimization Techniques

CO1.To imparts knowledge in concepts and tools of Operations Research.

CO2. To understand mathematical models used in Operations Research.

CO3. To apply these techniques constructively to make effective business decisions.

MCA-305 Elective I: EI(h) : Computer Graphics

CO1. To introduce the use of the components of a graphics system and become familiar with building approach of graphics system components and algorithms related with them.

CO2. To learn the basic principles of 3- dimensional computer graphics.

CO3. Provide an understanding of how to scan convert the basic geometrical primitives, how to transform the shapes to fit them as per the picture definition.

CO4. Provide an understanding of mapping from a world coordinates to device coordinates, clipping, and projections.

CO5. To be able to discuss the application of computer graphics concepts in the development of computer games, information visualization, and business applications.

CO6. To comprehend and analyze the fundamentals of animation, virtual reality, underlying technologies, principles, and applications.